Conserve mechanical energy

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Track kinetic and potential energy when nonconservative work is negligible.

SUCCESS CRITERIA - I can...

1. Identify the energy system and assumptions.
2. Keep the total energy consistent between positions.
3. Explain how height and speed trade off.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “mechanical energy” and “conserved” to explain your model or result. Sentence starters are optional.

mechanical energy: total kinetic and potential energy in a model

conserved: unchanged in total under the stated conditions

system: the object or objects chosen for analysis

01 Bring back an earlier idea

A 3 kg object descends 5 m near Earth.
Use g=10.
Find ΔU and the work done by gravity.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $K_i + U_i = K_f + U_f$; $K = \frac{1}{2}mv^2$; $U = mgh$.
Mechanical energy conserved when only conservative interactions transfer energy. Use g = 10 N/kg near the surface.

Worked reasoning: A 1 kg object drops 5 m from rest with g=10 m/s² and no energy loss. It loses 50 J of potential energy and gains 50 J of kinetic energy, giving speed 10 m/s.

Your turn: Draw energy bars at the start and just before arrival.
Label the system and neglected losses.

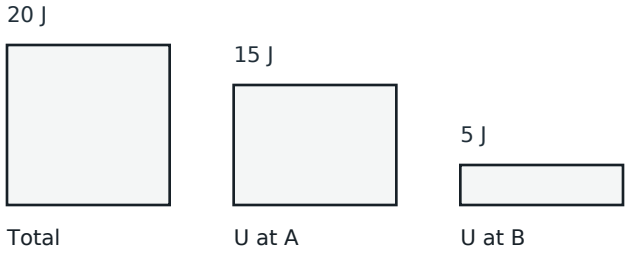
Sentence starter: The … in my model represents … . Its label shows … .

Marzano check - circle: ① ② ③

03 Represent and interpret

A system has total mechanical energy 20 J. At one point U=15 J; at another U=5 J.
Find K at each point.

Sentence starter: I found … [unit]. This value means … in this situation.



Bar height represents energy. Read the value and unit above each bar.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Solve and justify

A 2 kg object begins at height 5 m with speed 4 m/s.
Find speed at height 1 m with no drag.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

05 Analyze the reasoning

A cart loses mechanical energy on a rough track. Does total energy conservation fail?
Name the destination to include.

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: 1 2 3

06 Choose a method

A and B have equal momentum magnitude, but A has 4 times B’s mass.
Compare speeds and kinetic energies algebraically.

Word help: momentum = mass times velocity, including direction

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: 1 2 3

07 Exit check

A ball is thrown upward at 10 m/s from h=0.
Find maximum height without drag, using g=10.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A 2 kg cart starts at 4 m/s and rises 3 m without friction; g=10. Can it reach that height?
Find its maximum rise.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3